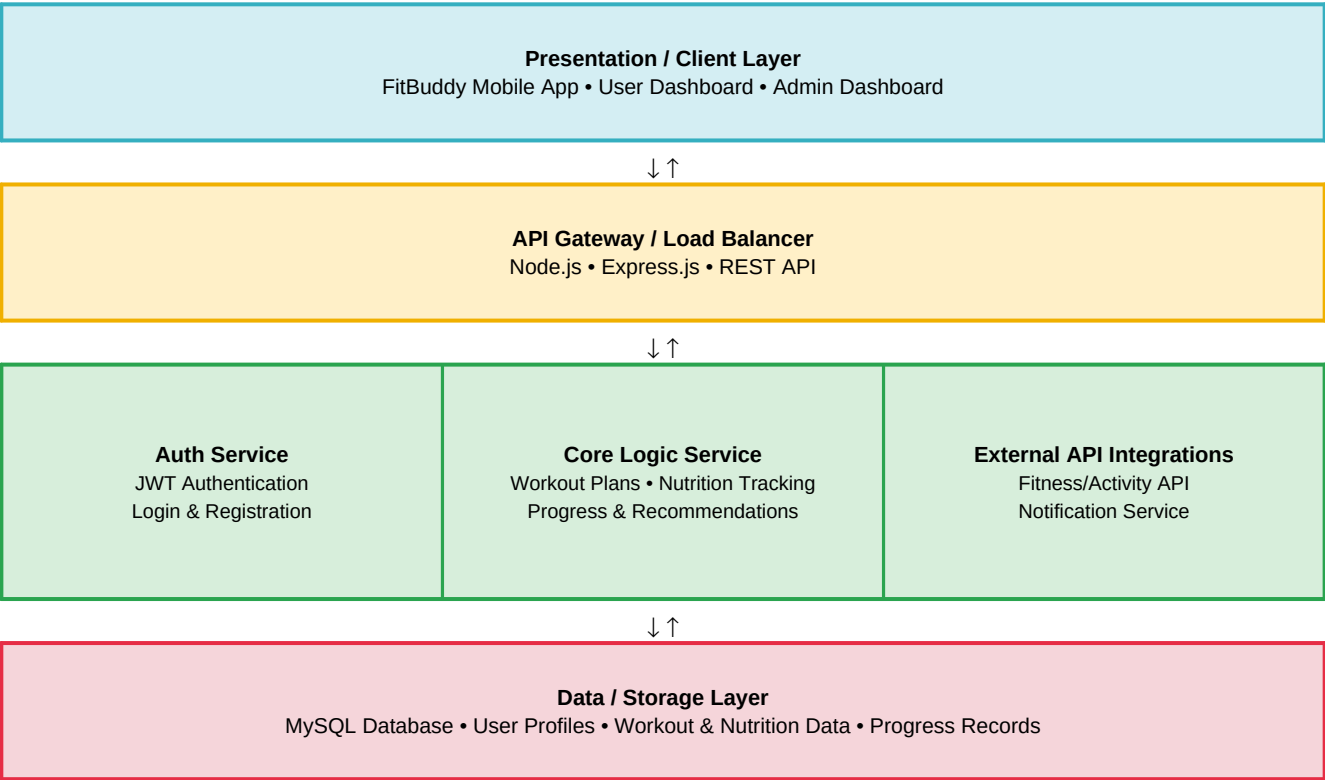


Solution Architecture

Date	30.09.2026
Team ID	SWTID-2026-7387
Project Name	FitBuddy
Maximum Marks	5 Marks

Solution Architecture Diagram:

High-level architecture for FitBuddy showing the client, application services, authentication, external integrations, and data storage layers.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the FitBuddy interface for viewing workouts, tracking nutrition, monitoring progress, and managing the user profile.	Flutter, Dart
API Gateway	Receives client requests, routes them to application services, and provides a common REST API entry point.	Node.js, Express.js, REST API
Microservices	Handles authentication, workout planning, nutrition tracking, progress monitoring, and personalized recommendations.	Node.js, JWT, Python/ML
Database	Stores user profiles, fitness goals, workout plans, nutrition records, activity data, and progress history.	MySQL